

Unit 7 Test 2 Review — Roots, Factors, Cubes, Grouping, & Rational Root Theorem

Name: _____

Date: _____

Period: _____

Sum: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

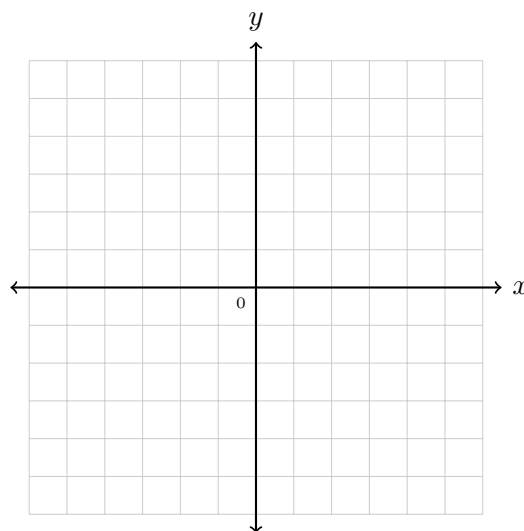
Diff: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

SOAP: Same, Opposite, Always Positive

Rational Root Theorem: possible roots = $\pm \frac{\text{factors of constant}}{\text{factors of leading coeff.}}$

Part 1: Roots and Factors (A2.7.D)

1. Find the real roots of $f(x) = x^3 - 6x^2 + 11x - 6$ and plot them on the grid below.



2. Let $f(x) = x^3 - 3x^2 - 13x + 15$. Which statements are true? **Select TWO.**

A. $x + 3$ is a linear factor of f

D. $x = -5$ is a zero of f

B. $f(3) = 0$

E. The remainder when $f(x) \div (x - 3)$ is zero

C. $(-3, 0)$ is an x -intercept of f

3. Determine if each is a factor of $f(x) = x^4 - 10x^2 + 9$.

	Factor	Not a Factor
$(x - 1)$		
$(x + 3)$		
$(x^2 + 9)$		
$(x - 3)$		
$(x + 1)$		
$(x^2 - 1)$		

4. Which is a factor of $y = x^4 - 625$?

A. $x^2 - 25$ C. Both are factors
 B. $x^2 + 25$ D. Neither are factors

5. Sam says $x^4 - 256 = (x^2 + 16)(x^2 - 16)$.
 Alex says $x^4 - 256 = (x^2 + 16)(x + 4)(x - 4)$.
 Who is correct? Explain.

Part 2: Factoring by Grouping (A2.7.D / A2.7.E) — Factor completely.

6. $2x^3 + x^2 - 50x - 25$

8. Factors of $x^3 + 7x^2 + 2x + 14$? **Select TWO.**

A. $x + 2$
 B. $x + 7$
 C. $x^2 + 7$
 D. $x^2 + 2$
 E. $x - 7$

7. Factors of $6x^3 - 9x^2 - 2x + 3$?

$2x - 3$	$3x + 1$	$3x - 1$
$2x^2 - 1$	$3x^2 - 1$	$3x^2 + 1$

Linear Factor: _____

Quadratic Factor: _____

Part 3: Sum & Difference of Cubes (A2.7.E)

9. Select TWO factors of $27x^3 + 8y^6$:

A. $3x + 2y^2$ B. $3x + 2y$ C. $3x - 2y^2$ D. $9x^2 - 6xy^2 + 4y^4$ E. $9x^2 + 6xy^2 + 4y^4$

10. Factor: $8x^3 - 27y^3$

11. Factor: $x^6 + 729$

(Hint: Rewrite as $(x^2)^3 + 9^3$)**Part 4: Rational Root Theorem (A2.7.D)**

12. $f(x) = 2x^3 - 5x^2 - 4x + 3$

a) List ALL possible rational roots: _____

b) Find one root using synthetic division, then c) factor $f(x)$ completely.

13. $g(x) = 3x^3 - x^2 - 12x + 4$

a) Possible rational roots: _____

b) Use Rational Root Theorem + synthetic division to factor $g(x)$ completely.**Part 5: Mixed Practice — Choose Your Strategy!**

14. $x^3 + 5x^2 - 9x - 45$

15. $216x^3 + 343$

16. $5x^3 - 15x^2 - 2x + 6$

Strategy: _____

Strategy: _____

Strategy: _____

BONUS: Factor $(x + 3)^3 - 64$ as a difference of cubes. Simplify each factor.

A. $(x-1)(x^2+10x+37)$ B. $(x-1)(x^2+6x+37)$ C. $(x+7)(x^2+10x+37)$ D. $(x-1)(x^2+10x+25)$